

Nonrephasing Controls for the SOC Bright–Dark Model

LiouvilleSolver SOC model notes

July 13, 2026

1 Purpose

This document compares the existing rephasing and nonrephasing third-order 1Q spectra for the four priority SOC-model tests. No new spectra were calculated here; the analysis reads the saved `S_data.npz` files and extracts finite-window observables.

The spectral maps below use independent colour normalisation for each panel. The quantitative amplitude comparison is therefore given by the tables, not by the visual colour scale.

For the spectral maps, rephasing is shown in the conventional $(-\omega_1, +\omega_3)$ quadrant, while nonrephasing is shown in the $(+\omega_1, +\omega_3)$ quadrant. The displayed maps use zoom spectra evaluated on a narrow window around the bright feature.

2 Validation summary

- Corrected spectra place rephasing in the $(-\omega_1, +\omega_3)$ quadrant and nonrephasing in the $(+\omega_1, +\omega_3)$ quadrant.
- The nonrephasing peak is centered at positive ω_1 for every displayed case.
- The corrected zoom-window amplitudes are comparable to the rephasing amplitudes in this symmetric model.
- The earlier broad-grid nonrephasing ratios should not be used for amplitude conclusions because they sampled $\omega_1 \neq 0$.

3 Step 6: dynamic phonon parameter scans

The nonrephasing channel tests whether the redistribution with g_Q and ω_Q survives a different time ordering.

Case	max NR/R	$ S_{NR} / S_R $	area NR/R	NR peak ω_1
gQ_0	1.018	1.018	1.018	1.004
gQ_0p01	1.037	1.041	1.048	1.005
omegaQ_0p07	1.051	1.054	1.073	1.006

Table 1: Finite-window nonrephasing ratios for Step 6: dynamic phonon parameter scans.

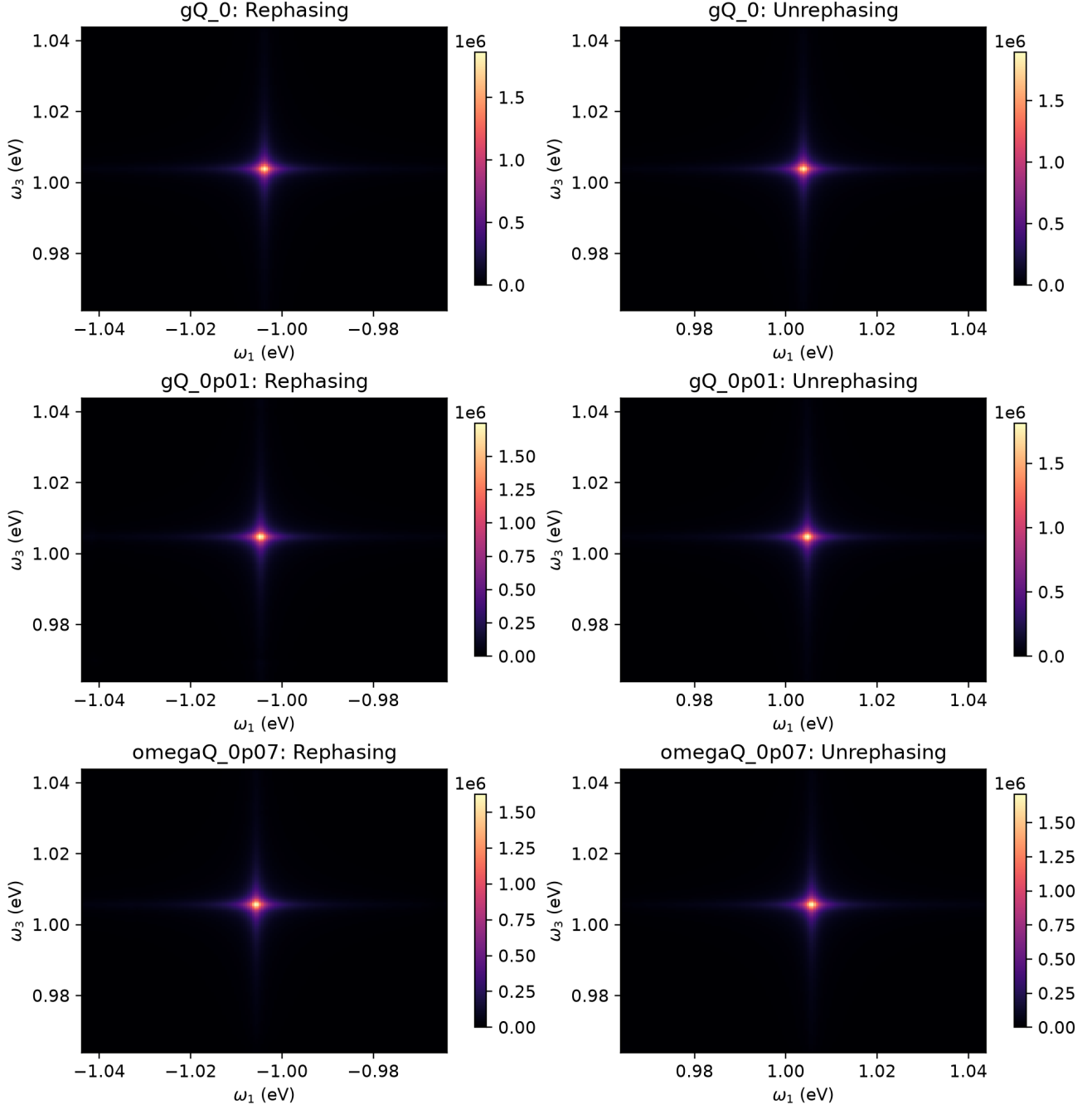


Figure 1: Representative $|S|$ spectra. Left column: rephasing; right column: nonrephasing. Each panel is independently normalised.

4 Step 3: static controls versus dynamic phonon

The key check is whether the two static controls remain degenerate while the dynamic phonon case becomes distinct.

Case	max NR/R	$ S_{NR} / S_R $	area NR/R	NR peak ω_1
static_dimerisation	1.018	1.018	1.018	1.004
dynamic_dimerisation_phonon	1.037	1.041	1.048	1.005

Table 2: Finite-window nonrephasing ratios for Step 3: static controls versus dynamic phonon.

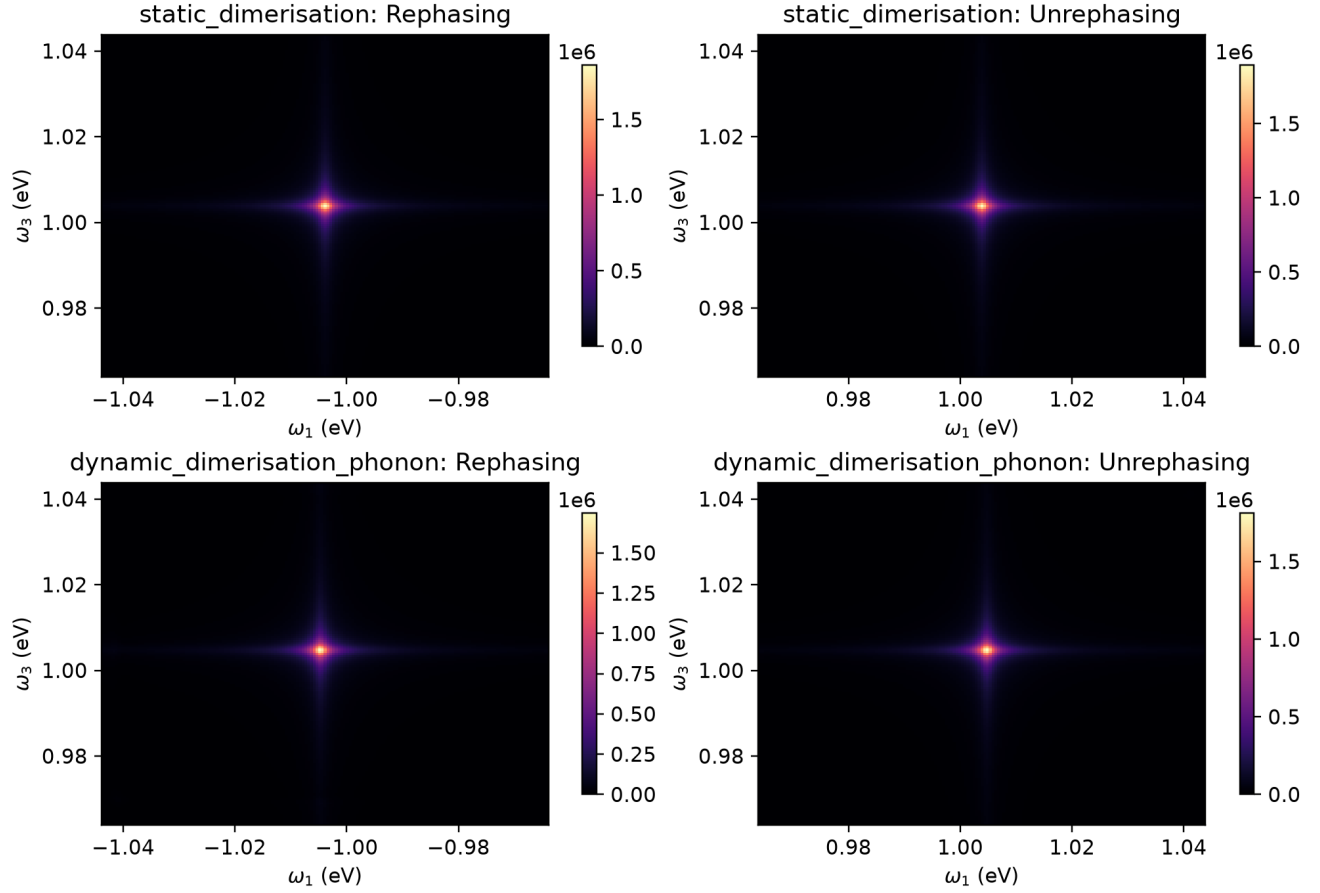


Figure 2: Representative $|S|$ spectra. Left column: rephasing; right column: nonrephasing. Each panel is independently normalised.

5 Step 5: temperature and field trends

The nonrephasing channel checks whether the lattice-off trend above T_{SP} or at high field is not a rephasing-only artefact.

Case	max NR/R	$ S_{NR} / S_R $	area NR/R	NR peak ω_1
T_lattice_T_7	1.037	1.041	1.048	1.005
T_lattice_T_20	1.005	1.005	1.005	1.001
T_C1_static_T_20	1.018	1.018	1.018	1.004

Table 3: Finite-window nonrephasing ratios for Step 5: temperature and field trends.

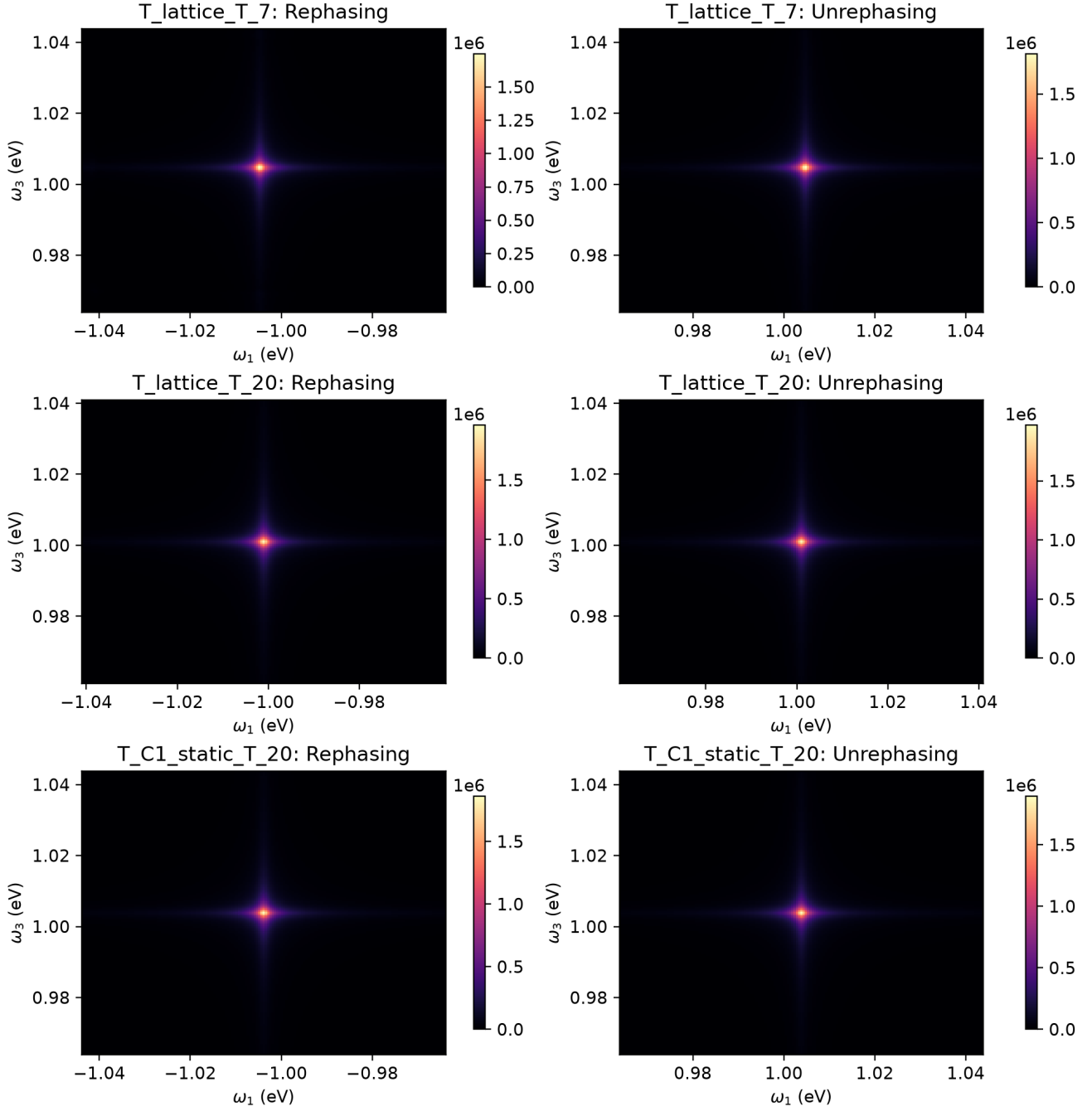


Figure 3: Representative $|S|$ spectra. Left column: rephasing; right column: nonrephasing. Each panel is independently normalised.

6 Step 2: scalar effective mixing

This is the lower-priority scalar control: if dimerisation and local C_1 enter only through the same V_{BD}^{eff} , nonrephasing should not separate them either.

Case	max NR/R	$ S_{NR} / S_R $	area NR/R	NR peak ω_1
dimerisation_control	1.018	1.018	1.018	1.004
spin_correlation_control	1.018	1.018	1.018	1.004

Table 4: Finite-window nonrephasing ratios for Step 2: scalar effective mixing.

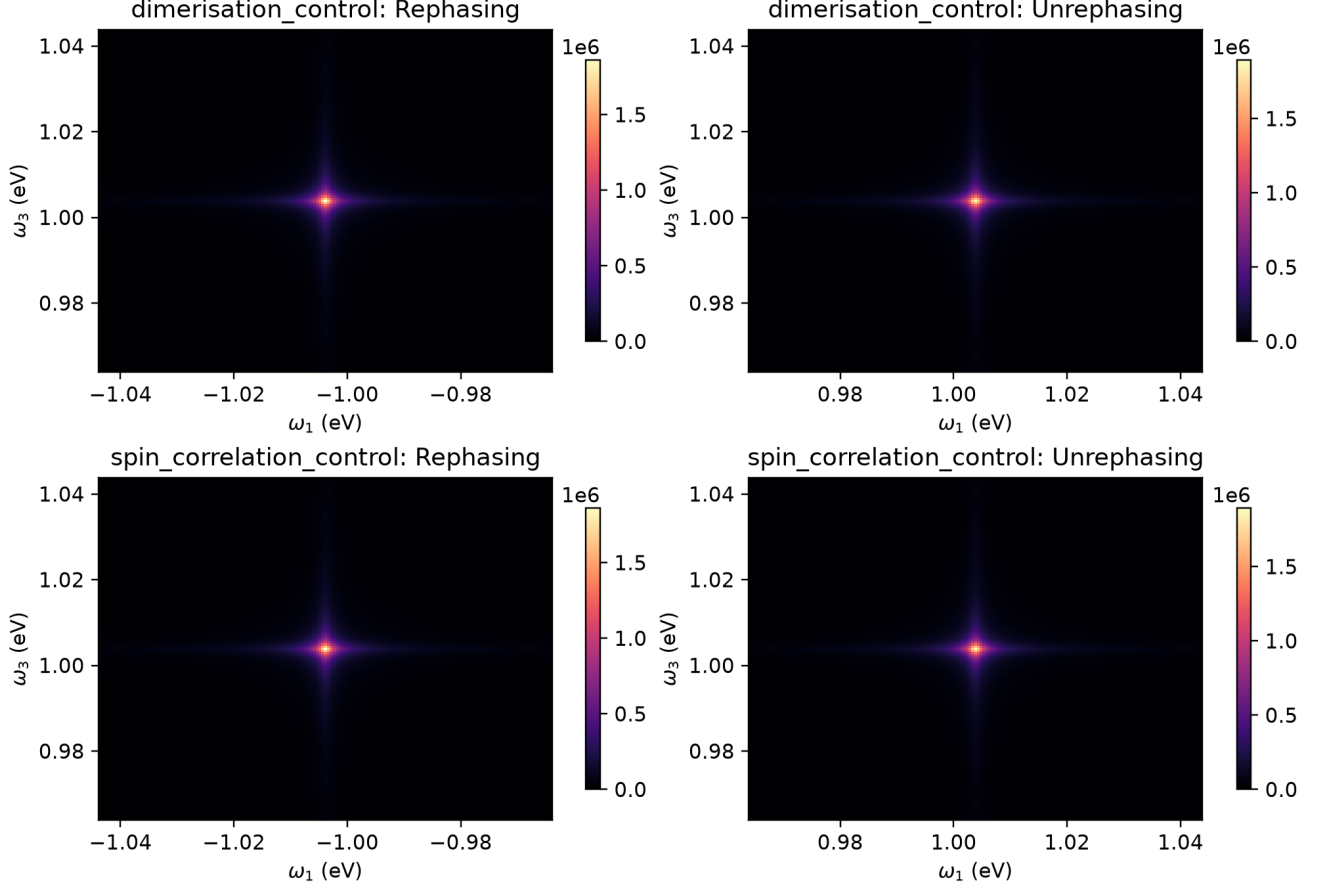


Figure 4: Representative $|S|$ spectra. Left column: rephasing; right column: nonrephasing. Each panel is independently normalised.

7 Overall conclusion

After evaluating nonrephasing in the correct positive-coherence quadrant, the displayed nonrephasing peaks are comparable in strength to the rephasing peaks for this symmetric model. The useful role of the nonrephasing channel is therefore not that it is a small correction, but that it provides a distinct pathway-order and quadrant check. It confirms the scalar degeneracy of Step 2, preserves the static-control degeneracy while separating the dynamic phonon case in Step 3, and gives a corrected basis for testing the trend controls in Step 5 and the dynamic phonon scans in Step 6.